

The light bulb

Scientists spent much of the 19th century trying to turn electricity into light. Their goal was to create a long-lasting electric light for home use. Two inventors—Joseph Swan and Thomas Edison—working independently on either side of the Atlantic Ocean, came up with the solution: the incandescent light bulb. This invention transformed the world.

LIGHTING THE WAY

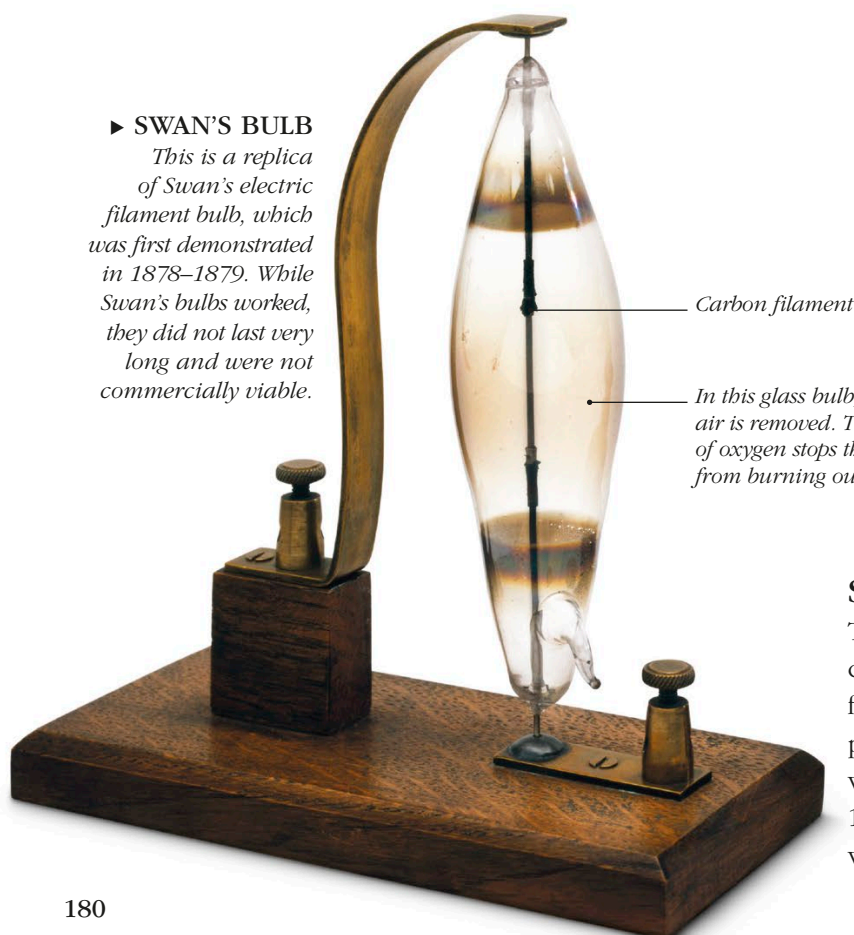
Invented by the British scientist Humphry Davy in 1809, the arc lamp worked by passing a bright arc of electricity through the air between two charcoal rods—like a controlled lightning strike. Although it was the first practical light, it was too powerful for home use. Arc lamps were used mainly for street lighting.



Arc lamps illuminate a street in New York in 1881

► SWAN'S BULB

This is a replica of Swan's electric filament bulb, which was first demonstrated in 1878–1879. While Swan's bulbs worked, they did not last very long and were not commercially viable.



Carbon filament

In this glass bulb, most of the air is removed. The absence of oxygen stops the filament from burning out.



LIFE IN THE DARK

Before the invention of the electric light, the world was a much darker place. People lit up their homes using candles made of animal fat and beeswax, as well as oil lamps and gas lamps. All were much less effective than electricity—a 100-watt light bulb is more than 100 times brighter than a candle flame.

WOW!

In the US, sales of light bulbs rocketed—from roughly 300,000 in 1885 to 795 million in 1945.

SWAN'S LIGHT

The British scientist Joseph Swan created the first light bulb suitable for the home, using a filament (a thin piece of material that glows brightly when electricity passes through it). In 1879, his home became the first in the world to be lit by electric light bulbs.